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System and method for individual player and team simulation

Abstract

A computing system retrieves historical event data for a plurality of games in a league. The historical event data includes (x,y) coordinates of players within each game and game context data. The computing system learns one or more attributes of each team in each game and each player on each team in each game. The computing system receives a request to simulate a play in a historical game. The request includes substituting a player that was in the play with a target player that was not in the play. The computing system simulates the play with the target player in place of the player based on the one or more attributes learned by the computing system. The computing system generates a graphical representation of the simulation.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6292706	12/2000	Birch et al.	N/A	N/A
6616529	12/2002	Qian et al.	N/A	N/A
6710713	12/2003	Russo	N/A	N/A
9342785	12/2015	Lucey et al.	N/A	N/A
9442933	12/2015	Tzoukermann et al.	N/A	N/A
9775396	12/2016	Olivares	N/A	N/A
10201752	12/2018	Lucey et al.	N/A	N/A
11157742	12/2020	Zhang et al.	N/A	N/A
2002/0165697	12/2001	Min	N/A	N/A
2004/0148278	12/2003	Milo et al.	N/A	N/A
2005/0143198	12/2004	Charge	N/A	N/A
2006/0149674	12/2005	Cook et al.	N/A	N/A
2006/0246972	12/2005	Thomas et al.	N/A	N/A
2006/0247808	12/2005	Robb	N/A	N/A
2006/0252476	12/2005	Bahou	N/A	N/A
2007/0265043	12/2006	Wang et al.	N/A	N/A
2007/0293289	12/2006	Loeb	N/A	N/A
2008/0281444	12/2007	Krieger et al.	N/A	N/A
2009/0186679	12/2008	Irvine et al.	N/A	N/A
2009/0203447	12/2008	Hansen et al.	N/A	N/A
2010/0184495	12/2009	Levy et al.	N/A	N/A
2010/0184563	12/2009	Molyneux et al.	N/A	N/A
2010/0283630	12/2009	Alonso	N/A	N/A
2011/0013087	12/2010	House et al.	N/A	N/A
2011/0013836	12/2010	Gefen et al.	N/A	N/A
2011/0169959	12/2010	DeAngelis et al.	N/A	N/A
2011/0173235	12/2010	Aman et al.	N/A	N/A
2011/0267461	12/2010	Birenboim et al.	N/A	N/A
2012/0046093	12/2011	Yamaguchi	N/A	N/A
2012/0162435	12/2011	Elangovan et al.	N/A	N/A
2012/0233105	12/2011	Cavallaro et al.	N/A	N/A
2012/0295677	12/2011	Ok	N/A	N/A

2013/0104870	12/2012	Rizzo et al.	N/A	N/A
2013/0110271	12/2012	Fornell et al.	N/A	N/A
2013/0225271	12/2012	Amaitis	N/A	N/A
2013/0267328	12/2012	Heisler et al.	N/A	N/A
2014/0142921	12/2013	Gleadall et al.	N/A	N/A
2014/0143183	12/2013	Sigal et al.	N/A	N/A
2014/0236331	12/2013	Lehmann et al.	N/A	N/A
2014/0248953	12/2013	Kelly et al.	N/A	N/A
2014/0274245	12/2013	Stickel	N/A	N/A
2014/0302914	12/2013	Weinstein et al.	N/A	N/A
2014/0309000	12/2013	Gustafson	N/A	N/A
2014/0364976	12/2013	Wohl et al.	N/A	N/A
2015/0131845	12/2014	Forouhar et al.	N/A	N/A
2015/0142716	12/2014	Lucey et al.	N/A	N/A
2015/0148129	12/2014	Austerlade et al.	N/A	N/A
2015/0273345	12/2014	Ford	N/A	N/A
2016/0182415	12/2015	Ames et al.	N/A	N/A
2016/0203279	12/2015	Srinivas et al.	N/A	N/A
2016/0220878	12/2015	Devathi	N/A	N/A
2016/0260015	12/2015	Lucey et al.	N/A	N/A
2016/0375365	12/2015	Thompson et al.	N/A	N/A
2017/0043260	12/2016	Austerlade et al.	N/A	N/A
2017/0061314	12/2016	Schnurr et al.	N/A	N/A
2017/0072321	12/2016	Thompson et al.	N/A	N/A
2017/0080336	12/2016	Groset et al.	N/A	N/A
2017/0109015	12/2016	Krasadakis	N/A	N/A
2017/0165570	12/2016	Lucey et al.	N/A	N/A
2017/0177930	12/2016	Holohan	N/A	N/A
2017/0235848	12/2016	Van Dusen et al.	N/A	N/A
2017/0238055	12/2016	Chang et al.	N/A	N/A
2017/0246539	12/2016	Schwartz et al.	N/A	N/A
2017/0255830	12/2016	Chen et al.	N/A	N/A
2017/0257653	12/2016	Farre et al.	N/A	N/A
2017/0291093	12/2016	Janssen et al.	N/A	N/A
2017/0330029	12/2016	Turcot et al.	N/A	N/A
2018/0032858	12/2017	Lucey et al.	N/A	N/A
2018/0054659	12/2017	Goswami et al.	N/A	N/A
2018/0084310	12/2017	Katz et al.	N/A	N/A
2018/0137364	12/2017	Forouhar et al.	N/A	N/A
2018/0218243	12/2017	Felsen et al.	N/A	N/A
2019/0087661	12/2018	Lee et al.	N/A	N/A
2019/0091585	12/2018	Amaitis	N/A	A63F 13/828
2019/0228306	12/2018	Power et al.	N/A	N/A
2019/0228316	12/2018	Felsen et al.	N/A	N/A
2019/0251366	12/2018	Zhong et al.	N/A	N/A
2020/0170549	12/2019	Baykaner et al.	N/A	N/A
2020/0218902	12/2019	Chang et al.	N/A	N/A
2020/0276474	12/2019	Power et al.	N/A	N/A
2020/0353311	12/2019	Ganguly et al.	N/A	N/A
2021/0241145	12/2020	Power et al.	N/A	N/A

2022/0108111	12/2021	Patton et al.	N/A	N/A
2022/0339538	12/2021	Lucey	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
1386241	12/2001	CN	N/A
105833502	12/2015	CN	N/A
100986647	12/2009	KR	N/A
2014008134	12/2013	WO	N/A
2015076682	12/2014	WO	N/A
2017031356	12/2016	WO	N/A
2017161167	12/2016	WO	N/A

OTHER PUBLICATIONS

International Search Report and Written Opinion for PCT Application No. PCT/US2019/014608, dated Apr. 15, 2019, 11 pages. cited by applicant

International Search Report and Written Opinion for PCT Application No. PCT/US2019/014609, dated Apr. 12, 2019, 10 pages. cited by applicant

International Search Report and Written Opinion for PCT Application No. PCT/US2019/014613, dated Apr. 12, 2019, 9 pages. cited by applicant

International Search Report and Written Opinion for PCT Application No. PCT/US2019/014614, dated Apr. 16, 2019, 8 pages. cited by applicant

Jain, et al., "Recurrent Neural Networks for Driver Activity Anticipation via Sensory-Fusion Architecture", IEEE International Conference on Robotics and Automation (ICRA), 2016, 8 pages. cited by applicant

Kharrat, et al., "Plus-Minus Player Ratings for Soccer", arXiv:1706.04943v1, Dated Jun. 16, 2017, 17 pages. cited by applicant

Kim, et al., "Motion Fields to Predict Play Evolution in Dynamic Sport Scenes", IEEE Computer Society Conference on Computer Vision and Pattern Recognition, 2010, 8 pages. cited by applicant

Kingma, et al., "Adam: A Method for Stochastic Optimization", arXiv:1412.6980v9, ICLR 2015, Dated Jan. 30, 2017, 15 pages. cited by applicant

Kingma, et al., "Auto-Encoding Variational Bayes", arXiv preprint arXiv:1312.6114, May 1, 2014, 14 pages. cited by applicant

Kingma, et al., "Semi-supervised Learning with Deep Generative Models", Advances in neural information processing systems, Oct. 31, 2014, 9 pages. cited by applicant

Kinoshita, et al., "Deep mixture density network for statistical model-based feature enhancement," 2017 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), Mar. 2017, pp. 251-255. cited by applicant

Knauf, et al., "Spatio-temporal convolution kernels", Machine learning 102.2, Jul. 2015, 28 pages. cited by applicant

Koopman, et al., "A dynamic bivariate Poisson model for analysing and forecasting match results in the English Premier League", Dated Sep. 24, 2012, 30 pages. cited by applicant

Koren, et al., "Matrix Factorization Techniques For Recommender Systems", Computer, IEEE Computer Society, Aug. 2009, 8 pages. cited by applicant

Le, et al., "Coordinated Multi-Agent Imitation Learning", Proceedings of the 34th International Conference on Machine Learning—vol. 70. JMLR.org, 2017, 13 pages. cited by applicant

Le, et al., "Data-Driven Ghosting using Deep Imitation Learning", MIT Sloan Sports Analytics Conference, Mar. 3-4, 2017, 15 pages. cited by applicant

Learned-Miller, "Data Driven Images Models through Continuous Joint Alignment", IEEE Transactions on Pattern Analysis and Machine Intelligence, vol. 28, No. 2, Feb. 2006, pp. 236-250.

cited by applicant

Lee, et al., “Desire: Distant Future Prediction in Dynamic Scenes with Interacting Agents”, Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2017, 10 pages. cited by applicant

Liang, et al., “Dual motion gan for future-flow embedded video prediction,” proceedings of the IEEE international conference on computer vision, 2017, 9 pages. cited by applicant

Lock, et al., “Using random forests to estimate win probability before each play of an NFL game”, JQAS 2014; 10(2); Published 2014, pp. 197-205. cited by applicant

Lucey, et al., “Quality vs Quantity: Improved Shot Prediction in Soccer using Strategic Features from Spatiotemporal Data”, 9th Annual MIT Sloan Sports Analytics Conference, Feb. 27-28, 2015, 9 pages. cited by applicant

Lucey, et al., “Representing and Discovering Adversarial Team Behaviors using Player Roles”, Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2013, 8 pages. cited by applicant

Maaten, et al., “Visualizing data using t-SNE”, Journal of machine learning research Sep. 2008, pp. 2579-2605, Nov. 2008. cited by applicant

Machine Learning Notebook, “Convolutional Neural Networks—Basics”, <https://mlnotebook.github.io/post/CNN1/>, Apr. 7, 2017, 13 pages. cited by applicant

Maher, “Modelling Association Football Scores”, Statistica Neerlandica 36, nr.3, Published 1982, pp. 109-118. cited by applicant

Maksai, et al., “What Players do with the Ball: A Physically Constrained Interaction Modeling”, Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2016, 10 pages. cited by applicant

McCabe, et al., “Artificial Intelligence in Sports Prediction”, Fifth International Conference on Information Technology: New Generation, IEEE Computer Society, Apr. 9, 2008, 4 pages. cited by applicant

Mehrotra, et al., “Elements of Artificial Neural Networks,” MIT, 1997, 351 pages. cited by applicant

Mikolov, et al., “Distributed Representations of Words and Phrases and their Compositionality”, Published 2013, 9 pages. cited by applicant

Mikolov, et al., “Efficient Estimation of Word Representations in Vector Space”, arXiv:1301.3781v3, Dated Sep. 7, 2013, 12 pages. cited by applicant

Miller, et al., “Factorized Point Process Intensities: A Spatial Analysis of Professional Basketball”, ICML, Jan. 8, 2014, 13 pages. cited by applicant

Newton, “How Youtub Perfected The Feed”, The Verge, <https://www.theverge.com/2017/8/30/16222850/youtube-google-brain-algorithm-video-recommendation-personalized-feed>, Aug. 30, 2017. cited by applicant

Paine, “The Pats' Comeback Was Incredible—Even if You Think the Falcons Blew it”, FiveThirtyEight, <https://fivethirtyeight.com/features/patriots-falcons-comeback-super-bowl/>, Dated Feb. 6, 2017, 4 pages. cited by applicant

Pathak, et al., “Contect Encoders: Feature Learning by Inpainting”, Proceedings of the IEEE conference on computer vision and pattern recognition, 2016, 12 pages. cited by applicant

PCT International Application No. PCT/US21/16583, International Search Report and Written Opinion of the International Searching Authority, dated Apr. 23, 2021, 12 pages. cited by applicant

PCT International Application No. PCT/US21/53117, International Search Report and Written Opinion of the International Searching Authority, dated Jan. 14, 2022, 9 pages. cited by applicant

PCT International Application No. PCT/US22/26534, International Search Report and Written Opinion of the International Searching Authority, dated Aug. 2, 2022, 8 pages. cited by applicant

Pelechrinis, “iWinRNFL: A Simple, Interpretable & Well-Calibrated In-Game Win Probability Model for NFL”, arXiv:1704.00197v3, Dated Mar. 14, 2018, 7 pages. cited by applicant

Peng, et al., "RASL: Robust Alignment by Sparse and Low-rank Decomposition for Linearly Correlated Images", PAMI, vol. 34, No. 11, 2012, 8 pages. cited by applicant

Perricone, et al., "Predicting Results for Professional Basketball Using NBA API Data," 2016, 6 pages. cited by applicant

Power, et al., "Not all passes are created equal: Objectively measuring the risk and reward of passes in soccer from tracking data", KDD 2017 Applied Data Science Paper, Aug. 2017, pp. 1605-1613. cited by applicant

Preferred Networks, Inc., "Football Analytics using Deep Learning", YouTube, https://www.youtube.com/watch?v=hs_v3dv6OUI, Jul. 25, 2019, 3 pages. cited by applicant

Pulling, "Long Corner Kicks in the English Premier League: Deliveries Into the Goal Area and Critical Area", Kinesiology: International journal of fundamental and applied kinesiology 47.2, 2015, pp. 193-201. cited by applicant

Ross, et al., "Shortcomings in the attribution process: On the origins and maintenance of erroneous social assessments", Cambridge University Press, 1982, pp. 129-152. cited by applicant

Ruiz, et al., "'The Leicester City Fairytale?' Utilizing New Soccer Analytics Tools to Compare Performance in the 15/16 & 16/17 EPL Seasons", KDD, Aug. 2017, 13 pages. cited by applicant

Schechtman-Rock, "Introducing NFLWin: An Open Source Implementation of NFL Win Probability", PhD Football, <http://phdfootball.blogspot.com/2016/09/introducing-nflwin-open-source.html>, Dated Sep. 1, 2016, 4 pages. cited by applicant

Seidl, et al., "Bhostgusters: Realtime interactive play sketching with synthesized NBA defenses", Proceeding of the 12th MIT Sloan Sports Analytics Conference, 2018. cited by applicant

Sha, et al., "Chalkboarding: A New Spatiotemporal Query Paradigm for Sports Play Retrieval", IUI, 2016, 12 pages. cited by applicant

Sha, et al., "Fine-Grained Retrieval of Sports Plays using Tree-Based Alignment of Trajectories", arXiv preprint arXiv:1710.02255, Oct. 6, 2017, 10 pages. cited by applicant

Sohn, et al., "Learning Structured Output Representation using Deep Conditional Generative Models", Advances in neural information processing systems, 2015, 9 pages. cited by applicant

Abdullah, et al., "Intelligent Prediction of Soccer Technical Skill on Youth Soccer Player's Relative Performance Using Multivariate Analysis and Artificial Neural Network Techniques", International Journal on Advanced Science Engineering Information Technology, vol. 6, No. 5, Dec. 31, 2016, pp. 668-674. cited by applicant

Acuna, "Unsupervised modeling of the movement of basketball players using a deep generative model," http://www.cs.toronto.edu/~davidj/projects/unsupervised_modeling_using_a_DGM.pdf, 31st Conference on Neural Information Processing Systems, 2017, 11 pages. cited by applicant

Akhter, et al., "Bilinear Spatiotemporal Basis Models", ACM Transactions on Graphics, vol. 30, No. 2, Article 17, Apr. 2012, 12 pages. cited by applicant

Alahi, et al., "Social LSTM: Human Trajectory Prediction in Crowded Spaces", Proceedings of the IEEE conference on computer vision and pattern recognition, 2016, 11 pages. cited by applicant

Alcorn, "(batter|pitcher) 2vec: statistic-free talent modeling with neural player embeddings", MIT Sloan Sports Analytics Conference, 2018. cited by applicant

Al-Shboul, et al., "Automated Player Selection for Sports Team using Competitive Neural Networks", International Journal of Advanced Computer Science and Applications (IJACSA), vol. 8, No. 8, 2017, pp. 457-460. cited by applicant

Anderson, et al., "The Numbers Game: Why Everything You Know About Soccer is Wrong", 2013, Penguin Books. cited by applicant

Arabzad, et al., "Football Match Results Prediction using Artificial Neural Networks; The case of Iran Pro League," Journal of Applied Research on Industrial Engineering, vol. 1, No. 3, Oct. 9, 2014, pp. 159-179. cited by applicant

Asif, et al., "In-Play forecasting of win probability in One-Day International cricket: A dynamic logistic regression model", International Journal of Forecasting 32, Published 2016, pp. 34-43.

cited by applicant

Bengio, et al., “A Neural Probabilistic Language Model”, Journal of Machine Learning Research 3, Published 2003, pp. 1137-1155. cited by applicant

Beuoy, “Updated NBA Win Probability Calculator”, InPredictable, <http://www.inpredictable.com/2015/02/updated-nba-win-probability-calculator.html>, Dated Feb. 6, 2015, 4 pages. cited by applicant

Bialkowski, et al., “Large-Scale Analysis of Soccer Matches using Spatiotemporal Tracking Data”, ICDM, 2014, 6 pages. cited by applicant

Bialkowski, et al., “Large-Scale Analysis of Soccer Matches using Spatiotemporal Tracking Data”, IEEE International Conference on Data Mining, 2014, 6 pages. cited by applicant

Bishop, “Mixture Density Networks”, Neural Computing Research Group Report, NCRG/94/004, Dated Feb. 1994, 26 pages. cited by applicant

Blaikie, et al. “NFL & NCAA Football Prediction using Artificial Neural network”, Proceedings of the Midstates Conference for Undergraduate Research in Computer Science and Mathematics, 2011, 8 pages. cited by applicant

Blei, et al., “Latent Dirichlet Allocation”, Journal of Machine Learning Research 3, Published 2003, pp. 993-1022. cited by applicant

Boice, “How Our Club Soccer Projections Work”, FiveThirtyEight, <https://fivethirtyeight.com/features/how-our-club-soccer-projections-work/>, Dated Jan. 19, 2017, 6 pages. cited by applicant

Boshnakov, et al., “A Bivariate Weibull Count Model For Forecasting Association Football Scores”, Dated Sep. 9, 2016, 13 pages. cited by applicant

Bowman, et al., “Generating Sentences from a Continuous Space”, arXiv preprint arXiv:1511.06349, May 12, 2016, 12 pages. cited by applicant

Butt, et al., “Multi-target Tracking by Lagrangian Relaxation to Min-Cost Network Flow”, Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2013, 8 pages. cited by applicant

Casal, et al., “Analysis of Corner Kick Success in Elite Football”, International Journal of Performance Analysis in Sport 2015, pp. 430-451. cited by applicant

Cervone, et al., “A multiresolution stochastic process model for predicting basketball possession outcomes”, Journal of the American Statistical Association 111.514, 2016, pp. 585-599. cited by applicant

Cervone, et al., “Pointwise: Predicting Points and Valuing Decisions in Real Time with NBA Optical Tracking Data”, 8th Annual MIT Sloan Sports Analytics Conference, Feb. 28-Mar. 1, 2014, 9 pages. cited by applicant

Chandler, et al., “An exploratory study of minor league baseball statistics,” Journal of Quantitative Analysis in Sports 8.4, 2012. cited by applicant

Charles, et al., “Personalizing Human Video Pose Estimation”, Proceedings of the IEEE conference on computer vision and pattern recognition, 2016, 14 pages. cited by applicant

Chen, et al., “Learning Online Smooth Predictors for Realtime Camera Planning using Recurrent Decision Trees”, Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2016, 14 pages. cited by applicant

Cheng, et al., “Wide & deep learning for recommender systems”, Proceedings of the 1st workshop on deep learning for recommender systems, Sep. 2016. cited by applicant

Chung, et al., “Hierarchical Multiscale Recurrent Neural Networks”, ICLR 2017, arXiv:1609.01704v7, Dated Mar. 9, 2017, 13 pages. cited by applicant

Covington, et al., “Deep neural networks for youtube recommendations”, Proceedings of the 10th ACM conference on recommender systems, pp. 191-198, Sep. 2016. cited by applicant

Cox, et al., “Least Squares Congealing for Unsupervised Alignment of Images”, CVPR, 2008, 8 pages. cited by applicant

Deerwester, et al., "Indexing by Latent semantic Analysis", Published 1990, 34 pages. cited by applicant

Deng, et al., "Factorized Variational Autoencoders for Modeling Audience Reactions to Movies", Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2017, 10 pages. cited by applicant

Dertat, "Applied deep learning-part 4: Convolutional neural networks", Towards Data Science, <https://towardsdatascience.com/applied-deep-learning-part-4-convolutional-neural-networks-584bc134c1e2>, Nov. 8, 2017, 26 pages. cited by applicant

Dixon, et al., "Modelling Association Football Scores and Inefficiencies in the Football Betting Market", Appl Statist 46. No 2, Published 1997, pp. 265-280. cited by applicant

Dubow, "San Antonio Spurs' Kawhi Leonard ruled out for Game 2 vs. Golden State Warriors", <http://www.nba.com/article/2017/05/15/san-antonio-spurs-kawhi-leonard-out-game-2>, Dated May 15, 2017, 4 pages. cited by applicant

Edge AI and Vision Alliance, "Using Convolutional Neural Networks for Image Recognition", <https://www.edge-ai-vision.com/2015/11/using-convolutional-neural-networks-for-image-recognition/>, Nov. 12, 2015, 20 pages. cited by applicant

ESPN Staff, "Neymar from Barca to PSG for €222m: Timeline of the world's biggest transfer", ESPN, <http://www.espn.com/soccer/blog/soccer-transfers/613/post/3170513/neymar-from-barcelona-to-psg-for-222m-euros-timeline-as-worlds-biggest-transfer-nears>, Aug. 4, 2017, 18 pages. cited by applicant

Felsen, et al., "Body shots: Analyzing shooting styles in the NBA using body pose", MIT Sloan, Sports Analytics Conference, Mar. 2017. cited by applicant

Felsen, et al., "What will Happen Next? Forecasting Player Moves in Sports Videos", Proceedings of the IEEE International Conference on Computer Vision, 2017, 10 pages. cited by applicant

Felsen, et al., "Where will they go? predicting fine-grained adversarial multi-agent motion using conditional variational autoencoders," Springer Nature Switzerland AG, Proceedings of the European conference on computer vision (ECCV), 2018, pp. 761-776. cited by applicant

Frey, et al., "Clustering by Passing Messages Between Data Points" Science Magazine, Feb. 15, 2007, 23 pages. cited by applicant

Goddard, et al., "Modelling football match results and the efficiency of fixed-odds betting", Published 2004, 27 pages. cited by applicant

Graves, "Generating Sequences with Recurrent Neural Networks", arXiv:1308.0850v5, Dated Jun. 5, 2014, 43 pages. cited by applicant

Gregor, et al., "DRAW: A Recurrent Neural Network For Image Generation", arXiv preprint arXiv:1502.04623, May 20, 2015, 10 pages. cited by applicant

Gyarmati, et al., "Searching for a unique style in soccer", KDD Workshop on Large-Scale Sports Analytics, 2014, 4 pages. cited by applicant

Hinton, et al. "A Fast Learning Algorithm for Deep Belief Nets", Neural Comput, 18(7), 2006, pp. 1527-1554. cited by applicant

Ho, "Random Decision Forests", Published 1995 IEEE, pp. 278-282. cited by applicant

Huang, et al., "A Neural Network Method for Prediction of 2006 World Cup Football game", Published 2010 IEEE, 8 pages. cited by applicant

Insafutdinov, et al., "ArtTrack: Articulated Multi-Person Tracking in the Wild", Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2017, 9 pages. cited by applicant

International Search Report and Written Opinion for PCT Application No. PCT/US2016/066733, dated Apr. 17, 2017, 11 pages. cited by applicant

Srinivas, et al., "A taxonomy of deep convolutional neural nets for computer vision", Frontiers in Robotics and AI 2, Jan. 11, 2016, 13 pages. cited by applicant

Su, et al., "Social Behavior Prediction from First Person Videos", arXiv preprint arXiv:1611.09464, Nov. 29, 2016, 10 pages. cited by applicant

Trainor, "Goalkeepers: How repeatable are shot saving performances?", STATSBOMB, <https://statsbomb.com/2014/10/goalkeepers-how-repeatable-are-shot-saving-performances/>, Oct. 21, 2014. cited by applicant

Van den Oord, et al., "Pixel Recurrent Neural Networks", arXiv preprint arXiv:1601.06759, Aug. 19, 2016, 11 pages. cited by applicant

Van den Oord, et al., "WaveNet: A Generative Model For Raw Audio", arXiv:1609.03499v2, Dated Sep. 19, 2016, 15 pages. cited by applicant

Walker, et al., "An Uncertain Future: Forecasting from Static Images using Variational Autoencoders", European Conference on Computer Vision, 2016, 17 pages. cited by applicant

Walters, "Virgil van Dijk transfer may not fix Liverpool's dismal defending of set-pieces warns Jurgen Klopp after Watford woe", <https://www.mirror.co.uk/sport/football/news/liverpool-klopp-van-dijk-setpiece-10981217>, Aug. 13, 2017, 21 pages. cited by applicant

Wang, et al., "Classifying NBA Offensive Plays Using Neural Networks," MIT Sloan Sports Analytics Conference, 2016, 9 pages. cited by applicant

Wang, et al., "Learning Optimal Parameters For Multi-target Tracking", International journal of computer vision 122.3, 2017, 13 pages. cited by applicant

Wei, et al., "Predicting Serves in Tennis using Style Priors", KDD, 2015, 9 pages. cited by applicant

Wei, et al., "The thin edge of the wedge: Accurately predicting shot outcomes in tennis using style and context priors", Proceedings of the 10th Annual MIT Sloan Sport Anal Conference, Mar. 2016. cited by applicant

Wei., "Modelling and predicting adversarial behaviour using large amounts of spatiotemporal data," Diss. Queensland University of Technology, 2016, 167 pages. cited by applicant

Wright, "Liverpool's set-piece problem exposed in 3-3 draw with Watford", <https://www.skysports.com/football/news/15117/10985307/liverpools-set-piece-problem-exposed-in-3-3-draw-with-watford>, Dec. 8, 2017, 9 pages. cited by applicant

Yamaguchi, et al., "Who are you with and Where are you going?", CVPR 2011. IEEE, 2011, 8 pages. cited by applicant

Yue, et al., "Learning Fine-Grained Spatial Models for Dynamic Sports Play Prediction", ICDM, 2014, 10 pages. cited by applicant

Zhao, "Research and Application of Online Learning of Formation Strategy Based on Neural Network, Information Technology Collection of China Masters' Dissertations Full Text Database," No. 2, 2014, pp. 38-40. cited by applicant

Zhao, et al., "Applying deep bidirectional LSTM and mixture density network for basketball trajectory prediction," Optik 158, Aug. 2017, pp. 266-272. cited by applicant

Zheng, et al., "Generating Long-term Trajectories Using Deep Hierarchical Networks", Advances in Neural Information Processing Systems, 2016, 9 pages. cited by applicant

Anonymous: "AlphaStar: Mastering the Real-Time Strategy Game StarCraft II", Jan. 24, 2019 (Jan. 24, 2019), pp. 1-7, XP055698054, Retrieved from the Internet: URL:<https://deepmind.com/blog/article/alphastar-mastering-real-time-strategy-game-starcraft-ii> [retrieved on May 25, 2020]. cited by applicant

Anonymous: "Drivatar(TM) in Forza Motorsport—Microsoft Research", Apr. 19, 2014 (Apr. 19, 2014), XP055205468, Retrieved from the Internet: URL:<https://web.archive.org/web/20140419214409/http://research.microsoft.com/en-us/projects/drivatar/forza.aspx> [retrieved on Jul. 30, 2015]. cited by applicant

De Almeida Rocha Daniel et al: "Simulating Human Behaviour in Games using Machine Learning", 2019 18th Brazilian Symposium On Computer Games and Digital Entertainment (SBGames), IEEE, Oct. 28, 2019 (Oct. 28, 2019), pp. 163-172, XP033670638, DOI: 10.1109/SBGAMES.2019.00030 [retrieved on Dec. 4, 2019]. cited by applicant

Miyashita Shohei et al: "Developing game AI agent behaving like human by mixing reinforcement

learning and supervised learning”, 2017 18th IEEE/Acis International Conference On Software Engineering, Artificial Intelligence, Networking and Parallel/Distributed Computing (SNPD), IEEE, Jun. 26, 2017 (Jun. 26, 2017), pp. 489-494, XP033146395, DOI: 10.1109/SNPD.2017.8022767 [retrieved on Aug. 29, 2017]. cited by applicant

Vinyals Oriol et al: “Grandmaster level in StarCraft II using multi-agent reinforcement learning”, Nature,, vol. 575, No. 7782, Oct. 30, 2019 (Oct. 30, 2019), pp. 350-354, XP036927623, DOI: 10.1038/S41586-019-1724-Z [retrieved on Oct. 30, 2019]. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of, and claims the benefit of priority to, U.S. application Ser. No. 17/660,980, filed Apr. 27, 2022, which claims priority to U.S. Provisional Application Ser. No. 63/180,168, filed Apr. 27, 2021, the entireties of each of which are incorporated herein by reference.

FIELD OF THE DISCLOSURE

(1) The present disclosure generally relates to system and method for individual player and team simulation using historical tracking data.

BACKGROUND

(2) In professional sports, fans and commentators alike incessantly argue about how players compare players from different generations. One of the more famous debates centers around that of LeBron James and Michael Jordan. Both players have dominated their respective generations and are among the two most mentioned players considered to be the “best of all time.” While fans and commentators can debate the issue, there is currently no mechanism to quantify or visualize a matchup between the two players.

SUMMARY

(3) In some embodiments, a method is disclosed herein. A computing system retrieves historical event data for a plurality of games in a league. The historical event data includes (x,y) coordinates of players within each game and game context data. The computing system learns one or more attributes of each team in each game and each player on each team in each game. The computing system receives a request to simulate a play in a historical game. The request includes substituting a player that was in the play with a target player that was not in the play. The computing system simulates the play with the target player in place of the player based on the one or more attributes learned by the computing system. The computing system generates a graphical representation of the simulation.

(4) In some embodiments, a non-transitory computer readable medium is disclosed herein. The non-transitory computer readable medium includes one or more sequences of instructions, which, when executed by a processor, causes a computing system to perform operations. The operations include retrieving, by the computing system, historical event data for a plurality of games in a league. The historical event data include player information within each game and game context data. The operations further include learning, by the computing system, one or more attributes of each team in each game and each player on each team in each game. The operations further include receiving, by the computing system, a request to simulate a play in a historical game. The request includes substituting a player that was in the play with a target player that was not in the play. The operations further include simulating, by the computing system, the play with the target player in

place of the player based on the one or more attributes learned by the computing system. The operations further include generating, by the computing system, a graphical representation of the simulation.

(5) In some embodiments, a system is disclosed herein. The system includes a processor and a memory. The memory has programming instructions stored thereon, which, when executed by the processor, causes the system to perform operations. The operations include retrieving historical event data for a plurality of games in a league. The historical event data include player information within each game and game context data. The operations further include learning, by a simulation module, one or more attributes of each team in each game and each player on each team in each game. The operations further include receiving a request to simulate a play in a historical game. The request includes substituting a player that was in the play with a target player that was not in the play. The operations further include simulating, by the simulation module, the play with the target player in place of the player based on the one or more attributes learned by the simulation module. The operations further include generating a graphical representation of the simulation.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) So that the manner in which the above recited features of the present disclosure can be understood in detail, a more particular description of the disclosure, briefly summarized above, may be had by reference to embodiments, some of which are illustrated in the appended drawings. It is to be noted, however, that the appended drawings illustrated only typical embodiments of this disclosure and are therefore not to be considered limiting of its scope, for the disclosure may admit to other equally effective embodiments.

(2) FIG. 1 is a block diagram illustrating a computing environment, according to example embodiments.

(3) FIG. 2 is a block diagram illustrating the organization computing system of FIG. 1, according to example embodiments

(4) FIG. 3 is a flow diagram illustrating a method of simulating a play, according to example embodiments.

(5) FIG. 4 illustrates an example graphical user interface according to example embodiments.

(6) FIG. 5A is a block diagram illustrating a computing device, according to example embodiments.

(7) FIG. 5B is a block diagram illustrating a computing device, according to example embodiments.

(8) To facilitate understanding, identical reference numerals have been used, where possible, to designate identical elements that are common to the figures. It is contemplated that elements disclosed in one embodiment may be beneficially utilized on other embodiments without specific recitation.

DETAILED DESCRIPTION

(9) One or more techniques provided herein provides a system and method for replaying a play in a given event by changing the players involved in the play and simulating what the alternative outcome would be. For example, one of the more infamous plays in sports' history involves Michael Jordan hitting the game-winning jump shot against the Utah Jazz over Byron Russell in game 6 of the 1998 NBA Finals. Using the one or more techniques provided herein, the present system is able to replace Byron Russell with another player, such as, but not limited to Larry Bird, Wilt Chamberlain, Magic Johnson, LeBron James, or Kawhi Leonard.

(10) To achieve such functionality, the present system may utilize historical tracking data. Such historical tracking data may include the (x, y) coordinates of each player during the course of the

game and/or body-pose information of each player during the course of the game. Further, to account for differences in errors, the present system may utilize one or more model adaptation techniques (e.g., transfer learning) to ensure that differences between eras may be normalized to increase the accuracy of the simulation.

(11) While the present discussion is provided in the context of both soccer and basketball, those skilled in the art readily understand that such functionality may be extended to other sports.

(12) FIG. 1 is a block diagram illustrating a computing environment **100**, according to example embodiments. Computing environment **100** may include tracking system **102**, organization computing system **104**, and one or more client devices **108** communicating via network **105**.

(13) Network **105** may be of any suitable type, including individual connections via the Internet, such as cellular or Wi-Fi networks. In some embodiments, network **105** may connect terminals, services, and mobile devices using direct connections, such as radio frequency identification (RFID), near-field communication (NFC), Bluetooth™, low-energy Bluetooth™ (BLE), Wi-Fi™, ZigBee™, ambient backscatter communication (ABC) protocols, USB, WAN, or LAN. Because the information transmitted may be personal or confidential, security concerns may dictate one or more of these types of connection be encrypted or otherwise secured. In some embodiments, however, the information being transmitted may be less personal, and therefore, the network connections may be selected for convenience over security.

(14) Network **105** may include any type of computer networking arrangement used to exchange data or information. For example, network **105** may be the Internet, a private data network, virtual private network using a public network and/or other suitable connection(s) that enables components in computing environment **100** to send and receive information between the components of environment **100**.

(15) Tracking system **102** may be positioned in a venue **106**. For example, venue **106** may be configured to host a sporting event that includes one or more agents **112**. Tracking system **102** may be configured to record the motions of all agents (i.e., players) on the playing surface, as well as one or more other objects of relevance (e.g., ball, referees, etc.). In some embodiments, tracking system **102** may be an optically-based system using, for example, a plurality of fixed cameras. For example, a system of six stationary, calibrated cameras, which project the three-dimensional locations of players and the ball onto a two-dimensional overhead view of the court may be used. In some embodiments, tracking system **102** may be a radio-based system using, for example, radio frequency identification (RFID) tags worn by players or embedded in objects to be tracked. Generally, tracking system **102** may be configured to sample and record, at a high frame rate (e.g., 25 Hz). Tracking system **102** may be configured to store at least player identity and positional information (e.g., (x, y) position) for all agents and objects on the playing surface for each frame in a game file **110**.

(16) Game file **110** may be augmented with other event information corresponding to event data, such as, but not limited to, game event information (pass, made shot, turnover, etc.) and context information (current score, time remaining, etc.).

(17) Tracking system **102** may be configured to communicate with organization computing system **104** via network **105**. Organization computing system **104** may be configured to manage and analyze the data captured by tracking system **102**. Organization computing system **104** may include at least a web client application server **114**, a pre-processing agent **116**, a data store **118**, and a simulation module **120**. Each of pre-processing agent **116** and simulation module **120** may be comprised of one or more software modules. The one or more software modules may be collections of code or instructions stored on a media (e.g., memory of organization computing system **104**) that represent a series of machine instructions (e.g., program code) that implements one or more algorithmic steps. Such machine instructions may be the actual computer code the processor of organization computing system **104** interprets to implement the instructions or, alternatively, may be a higher level of coding of the instructions that is interpreted to obtain the actual computer code.

The one or more software modules may also include one or more hardware components. One or more aspects of an example algorithm may be performed by the hardware components (e.g., circuitry) itself, rather as a result of the instructions.

(18) Data store **118** may be configured to store one or more game files **124**. Each game file **124** may include spatial event data and non-spatial event data. For example, spatial event data may correspond to raw data captured from a particular game or event by tracking system **102**. Non-spatial event data may correspond to one or more variables describing the events occurring in a particular match without associated spatial information. For example, non-spatial event data may correspond to each play-by-play event in a particular match. In some embodiments, non-spatial event data may be derived from spatial event data. For example, pre-processing agent **116** may be configured to parse the spatial event data to derive play-by-play information. In some embodiments, non-spatial event data may be derived independently from spatial event data. For example, an administrator or entity associated with organization computing system may analyze each match to generate such non-spatial event data. As such, for purposes of this application, event data may correspond to spatial event data and non-spatial event data.

(19) In some embodiments, each game file **124** may further include the home and away team box scores. For example, the home and away teams' box scores may include the number of team assists, fouls, rebounds (e.g., offensive, defensive, total), steals, and turnovers at each time, t , during gameplay. In some embodiments, each game file **124** may further include a player box score. For example, the player box score may include the number of player assists, fouls, rebounds, shot attempts, points, free-throw attempts, free-throws made, blocks, turnovers, minutes played, plus/minus metric, game started, and the like. Although the above metrics are discussed with respect to basketball, those skilled in the art readily understand that the specific metrics may change based on sport. For example, in soccer, the home and away teams' box scores may include shot attempts, assists, crosses, shots, and the like.

(20) Pre-processing agent **116** may be configured to process data retrieved from data store **118**. For example, pre-processing agent **116** may be configured to generate one or more sets of information that may be used to train simulation module **120**.

(21) Simulation module **120** may be configured to simulate play between various players or teams based on historical tracking data of the players and/or teams. To do so, simulation module **120** may analyze historical games associated with a given league. For example, simulation module **120** may utilize data generated by AutoSTATS to analyze historical game information.

(22) Once simulation module **120** generates or learns various data associated with the historical game information (e.g., body pose information, role information, trajectory information, and/or style information), simulation module **120** may be able to simulate behavior a game or a play between teams or players.

(23) In some embodiments, simulation module **120** may be able to simulate a game or play between teams or players even across generations. For example, simulation module **120** may be able to simulate a play between LeBron James and Michael Jordan, despite the fact that the two players have never played an NBA game against each other. To accurately simulate the performance, simulation module **120** may utilize a transfer learning process. In this manner, simulation module **120** may be able to transfer knowledge of a player in a first era to the course of play in a second era. In other words, simulation module **120** may account for differences of play between eras. In this manner, simulation module **120** can adapt players of different eras to predict how they would play in another era.

(24) Client device **108** may be in communication with organization computing system **104** via network **105**. Client device **108** may be operated by a user. For example, client device **108** may be a mobile device, a tablet, a desktop computer, or any computing system having the capabilities described herein. Users may include, but are not limited to, individuals such as, for example, subscribers, clients, prospective clients, or customers of an entity associated with organization

computing system **104**, such as individuals who have obtained, will obtain, or may obtain a product, service, or consultation from an entity associated with organization computing system **104**.

(25) Client device **108** may include at least application **138**. Application **138** may be representative of a web browser that allows access to a website or a stand-alone application. Client device **108** may access application **138** to access one or more functionalities of organization computing system **104**. Client device **108** may communicate over network **105** to request a webpage, for example, from web client application server **114** of organization computing system **104**. For example, client device **108** may be configured to execute application **138** to request a simulation from simulation module **120** and/or view a simulation generated by simulation module **120**.

(26) FIG. 2 is a block diagram **200** illustrating organization computing system **104**, according to example embodiments. As shown, organization computing system **104** includes repository **202** and one or more computer processors **204**.

(27) Repository **202** may be representative of any type of storage unit and/or device (e.g., a file system, database, collection of tables, or any other storage mechanism) for storing data. Further, repository **202** may include multiple different storage units and/or devices. The multiple different storage units and/or devices may or may not be of the same type or located at the same physical site. As shown, repository **202** includes at least simulation module **120**.

(28) Simulation module **120** may include at least body pose module **206**, role module **208**, trajectory engine **210**, style module **212**, macroanalysis module **214**, and microanalysis module **216**. Each of body pose module **206**, role module **208**, trajectory engine **210**, style module **212**, macroanalysis module **214**, and microanalysis module **216** may be comprised of one or more software modules. The one or more software modules are collections of code or instructions stored on a media (e.g., memory of organization computing system **104**) that represent a series of machine instructions (e.g., program code) that implements one or more algorithmic steps. Such machine instructions may be the actual computer code the processor of organization computing system **104** interprets to implement the instructions or, alternatively, may be a higher level of coding of the instructions that are interpreted to obtain the actual computer code. The one or more software modules may also include one or more hardware components. One or more aspects of an example algorithm may be performed by the hardware components (e.g., circuitry) itself, rather than as a result of the instructions.

(29) As shown, simulation module **120** may be configured to receive tracking data and/or event data from tracking system **102**. In some embodiments, the tracking data may be representative of raw (x, y) position information of each player on the playing surface during a course of the game. In some embodiments, event data may indicate various attributes of the game at various points, such as, but not limited to, at one or more points during the game: the players in the game, the players out of the game, the current score, the current time, player statistics at the current time, and the like. In some embodiments, the tracking data may be derived from broadcast video data received from tracking system **102**.

(30) Body pose module **206** may be configured to generate one or more metrics related to the body pose of at least one or more of a player throughout a game based on the tracking data and/or event data. In some embodiments, body pose module **206** may generate body pose information based on event data captured by tracking system **102**. In some embodiments, body pose module **206** may generate body pose information from a broadcast stream provided by a broadcast provider. Body pose module **206** may be able to identify, for example, shooter start position and angle, run type (e.g., stutter and speed), shot initiation (e.g., body lean angle, upper body angle, hip orientation, kicking arm position, shoulder alignment, etc.), and the like. Additionally, the raw positions of the body-positions in 2D or 3D which appear as a skeleton can be used to detect and correlate specific key actions in sports. Generally, body pose module **206** may be representative of a body pose module disclosed in U.S. application Ser. No. 16/804,964, which is hereby incorporated by

reference in its entirety.

(31) Role module **208** may be configured to detect a role in each player of each historical game based on the tracking data and/or event data. For example, role module **208** may be configured to predict various different aspects of team and player functions within a team. For example, role module **208** may utilize an ensemble of models that may work in conjunction to learn the role associated with a given player. In some embodiments, role module **208** may utilize event data to make such determination. Generally, role module **208** may be representative of role prediction platform disclosed in U.S. application Ser. No. 17/167,400, which is hereby incorporated by reference in its entirety.

(32) Trajectory engine **210** may be configured to learn trajectories of each player in each historical game based on the tracking data and/or event data. For example, trajectory engine **210** may be configured to predict the trajectory of one or more agents given one or more historical trajectory points. For example, given an agent's coordinates up to a time $t_{sub.q}$, trajectory engine **210** may use at least the agent's coordinates up to time $t_{sub.q}$ to predict the agent's coordinates up to time $t_{sub.f}$, where $t_{sub.f}$ is after $t_{sub.q}$, where q represents some end-time between (e.g., $(1 \dots n)$) and f represents some future time, after q (e.g., $(n+1)$). Generally, trajectory engine **210** may be representative of trajectory agent disclosed in U.S. application Ser. No. 16/254,037, which is hereby incorporated by reference in its entirety.

(33) Style module **212** may be configured to learn the various styles and content of each team in each historical game based on the tracking data and/or event data. Content of a given play may be referred to as the “what” of the play, independent of the exact specifics of how a team executes the play. In contrast, style may be referred to as the “how” of the play, which captures the various ways a given play can evolve. Generally, style module **212** may be representative of team prediction agent disclosed in U.S. application Ser. No. 16/870,170, which is hereby incorporated by reference in its entirety.

(34) Macroanalysis module **214** may be configured to perform one or more macro analyses based on one or more of the learned body pose information, the learned role information, the learned trajectory information, or the learned style information. For example, macroanalysis module **214** may be configured to utilize one or more of the learned body pose information, the learned role information, the learned trajectory information, or the learned style information to answer counterfactual questions, such as, determining where Michael Jordan would be drafted in an upcoming NBA draft based on Michael Jordan's collegiate statistics at the University of North Carolina. In such example, macroanalysis module **214** may leverage the prediction model disclosed in U.S. application Ser. No. 17/449,694, which is hereby incorporated by reference in its entirety.

(35) In another example, macroanalysis module **214** may be configured to utilize one or more of the learned body pose information, the learned role information, the learned trajectory information, or the learned style information to answer counterfactual questions, such as, determining the performance of Pele in the English Premier League. In such example, macroanalysis module **214** may leverage the prediction models disclosed in U.S. Provisional Application No. 63/201,898 and U.S. Provisional Application No. 63/267,062, which are both hereby incorporated by reference in their entireties.

(36) Microanalysis module **216** may be configured to perform one or more micro analyses based on one or more of the learned body pose information, the learned role information, the learned trajectory information, or the learned style information. For example, microanalysis module **216** may be configured to simulate a play or a game by replacing a player in the play or game with another player based on one or more of the learned body pose information, the learned role information, the learned trajectory information, or the learned style information. For example, microanalysis module **216** may leverage one or more graph neural networks disclosed in U.S. application Ser. No. 17/649,970, which is hereby incorporated by reference in its entirety, to model defensive behavior of the player.

(37) FIG. 3 is a flow diagram illustrating a method **300** of simulating a game or play, according to example embodiments. Method **300** may begin at step **302**.

(38) At step **302**, organization computing system **104** may retrieve historical event data for a plurality of games in a league. In some embodiments, the historical event data may include the (x, y) coordinates of all players at all points in the game. In some embodiments, the historical event data may include game context data, such as the time of each movement, score at each movement, and the like. Generally, the plurality of games may span several eras. Taking basketball for example, the plurality of games may span the 1960s to current day.

(39) At step **304**, organization computing system **104** may learn various attributes of the teams and players in each game. For example, simulation module **120** may learn various attributes of the teams and players in each game, such that simulation module **120** can accurately simulate performance between players and/or teams that may not have ever played each other or did not play each other given the current game context.

(40) At step **306**, organization computing system **104** may receive a request to simulate a game or play between two teams. In some embodiments, simulation module **120** may receive the request from client device **108** via application **138** executing thereon. In some embodiments, the request may include replacing one player in a game or player with a different player that is otherwise not involved in the current play of the game. For example, continuing with the above, Lebron James may replace Michael Jordan in the game winning play of Game 6 of the 1998 NBA Finals.

(41) At step **308**, organization computing system **104** may simulate the game or play. For example, simulation module **120** may simulate the play between Lebron James and Byron Russell based on learned body pose information, role information, trajectory information, playing style and playing style information. In some embodiments, simulation module **120** may further take into account the difference between eras. For example, simulation module **120** may adapt Lebron James' playing style to the NBA playing style of 1998.

(42) At step **310**, organization computing system **104** may generate a graphical representation of the simulation. For example, simulation module **120** may generate a visual corresponding to Lebron James defending Michael Jordan in the game winning play of the 1998 NBA Finals. In some embodiments, simulation module **120** may provide a visual indication of each player's body pose information.

(43) FIG. 4 illustrates an example graphical user interface (GUI) **400** according to example embodiments. As shown, graphical user interface **400** may correspond to a simulation of Lebron James guarding Michael Jordan in the game winning play of the 1998 NBA Finals. For example, as shown, rather than Byron Russell guarding Michael Jordan in the game winning play, a user has requested that Lebron James be guarding Byron Russell. As shown, GUI **400** may include a skeletal representation of Lebron James guarding a skeletal representation of Michael Jordan. Based on learned information about Lebron James (e.g., one or more of the learned body pose information, the learned role information, the learned trajectory information, or the learned style information), simulation module **120** may simulate the game winning play as if Lebron James was guarding Michael Jordan in that situation rather than Byron Russell.

(44) Although GUI **400** illustrates a scenario in which the defensive player is replaced with another player, those skilled in the art understand that, instead of replacing Byron Russell, a user may instead replace Michael Jordan with Lebron James. In this manner, based on learned information about Lebron James (e.g., one or more of the learned body pose information, the learned role information, the learned trajectory information, or the learned style information), simulation module **120** may simulate the game winning play as if Lebron James had possession against Byron Russell instead of Michael Jordan.

(45) FIG. 5A illustrates a system bus architecture of computing system **500**, according to example embodiments. System **500** may be representative of at least a portion of organization computing system **104**. One or more components of system **500** may be in electrical communication with each

other using a bus **505**. System **500** may include a processing unit (CPU or processor) **510** and a system bus **505** that couples various system components including the system memory **515**, such as read only memory (ROM) **520** and random access memory (RAM) **525**, to processor **510**. System **500** may include a cache of high-speed memory connected directly with, in close proximity to, or integrated as part of processor **510**. System **500** may copy data from memory **515** and/or storage device **530** to cache **512** for quick access by processor **510**. In this way, cache **512** may provide a performance boost that avoids processor **510** delays while waiting for data. These and other modules may control or be configured to control processor **510** to perform various actions. Other system memory **515** may be available for use as well. Memory **515** may include multiple different types of memory with different performance characteristics. Processor **510** may include any general purpose processor and a hardware module or software module, such as service 1 **532**, service 2 **534**, and service 3 **536** stored in storage device **530**, configured to control processor **510** as well as a special-purpose processor where software instructions are incorporated into the actual processor design. Processor **510** may essentially be a completely self-contained computing system, containing multiple cores or processors, a bus, memory controller, cache, etc. A multi-core processor may be symmetric or asymmetric.

(46) To enable user interaction with the computing system **500**, an input device **545** may represent any number of input mechanisms, such as a microphone for speech, a touch-sensitive screen for gesture or graphical input, keyboard, mouse, motion input, speech and so forth. An output device **535** may also be one or more of a number of output mechanisms known to those of skill in the art. In some instances, multimodal systems may enable a user to provide multiple types of input to communicate with computing system **500**. Communications interface **540** may generally govern and manage the user input and system output. There is no restriction on operating on any particular hardware arrangement and therefore the basic features here may easily be substituted for improved hardware or firmware arrangements as they are developed.

(47) Storage device **530** may be a non-volatile memory and may be a hard disk or other types of computer readable media which may store data that are accessible by a computer, such as magnetic cassettes, flash memory cards, solid state memory devices, digital versatile disks, cartridges, random access memories (RAMs) **525**, read only memory (ROM) **520**, and hybrids thereof.

(48) Storage device **530** may include services **532**, **534**, and **536** for controlling the processor **510**. Other hardware or software modules are contemplated. Storage device **530** may be connected to system bus **505**. In one aspect, a hardware module that performs a particular function may include the software component stored in a computer-readable medium in connection with the necessary hardware components, such as processor **510**, bus **505**, output device **535** (e.g., display), and so forth, to carry out the function.

(49) FIG. 5B illustrates a computer system **550** having a chipset architecture that may represent at least a portion of organization computing system **104**. Computer system **550** may be an example of computer hardware, software, and firmware that may be used to implement the disclosed technology. System **550** may include a processor **555**, representative of any number of physically and/or logically distinct resources capable of executing software, firmware, and hardware configured to perform identified computations. Processor **555** may communicate with a chipset **560** that may control input to and output from processor **555**. In this example, chipset **560** outputs information to output **565**, such as a display, and may read and write information to storage device **570**, which may include magnetic media, and solid state media, for example. Chipset **560** may also read data from and write data to storage device **575** (e.g., RAM). A bridge **580** for interfacing with a variety of user interface components **585** may be provided for interfacing with chipset **560**. Such user interface components **585** may include a keyboard, a microphone, touch detection and processing circuitry, a pointing device, such as a mouse, and so on. In general, inputs to system **550** may come from any of a variety of sources, machine generated and/or human generated.

(50) Chipset **560** may also interface with one or more communication interfaces **590** that may have

different physical interfaces. Such communication interfaces may include interfaces for wired and wireless local area networks, for broadband wireless networks, as well as personal area networks. Some applications of the methods for generating, displaying, and using the GUI disclosed herein may include receiving ordered datasets over the physical interface or be generated by the machine itself by processor 555 analyzing data stored in storage device 570 or storage device 575. Further, the machine may receive inputs from a user through user interface components 585 and execute appropriate functions, such as browsing functions by interpreting these inputs using processor 555. (51) It may be appreciated that example systems 500 and 550 may have more than one processor 510 or be part of a group or cluster of computing devices networked together to provide greater processing capability.

(52) While the foregoing is directed to embodiments described herein, other and further embodiments may be devised without departing from the basic scope thereof. For example, aspects of the present disclosure may be implemented in hardware or software or a combination of hardware and software. One embodiment described herein may be implemented as a program product for use with a computer system. The program(s) of the program product define functions of the embodiments (including the methods described herein) and can be contained on a variety of computer-readable storage media. Illustrative computer-readable storage media include, but are not limited to: (i) non-writable storage media (e.g., read-only memory (ROM) devices within a computer, such as CD-ROM disks readably by a CD-ROM drive, flash memory, ROM chips, or any type of solid-state non-volatile memory) on which information is permanently stored; and (ii) writable storage media (e.g., floppy disks within a diskette drive or hard-disk drive or any type of solid state random-access memory) on which alterable information is stored. Such computer-readable storage media, when carrying computer-readable instructions that direct the functions of the disclosed embodiments, are embodiments of the present disclosure.

(53) It will be appreciated to those skilled in the art that the preceding examples are exemplary and not limiting. It is intended that all permutations, enhancements, equivalents, and improvements thereto are apparent to those skilled in the art upon a reading of the specification and a study of the drawings are included within the true spirit and scope of the present disclosure. It is therefore intended that the following appended claims include all such modifications, permutations, and equivalents as fall within the true spirit and scope of these teachings.

Claims

1. A method, comprising: retrieving, by a computing system, historical event data for a plurality of sporting matches, wherein retrieving the historical event data comprises: receiving, by the computing system, broadcast video data corresponding to the plurality of sporting matches; deriving, by the computing system, player tracking data from the broadcast video data; augmenting, by the computing system, the player tracking data with the historical event data; generating, by the computing system, one or more attributes of a team and of at least one player on the team based on the historical event data; receiving, by the computing system, a request to simulate and view an alternative historical game outcome; simulating, by the computing system, the alternative historical game outcome based on the one or more attributes generated by the computing system; and outputting, by the computing system, a graphical representation of the simulation.
2. The method of claim 1, the method further comprising: generating, by the computing system, body pose data for the at least one player based on the historical event data.
3. The method of claim 2, wherein the simulating the alternative historical game outcome is based on the body pose data.
4. The method of claim 1, wherein the graphical representation includes at least one skeletal representation of the at least one player, and wherein the at least one skeletal representation is based on the one or more attributes.

5. The method of claim 1, wherein the one or more attributes include one or more player trajectories, one or more team styles, and one or more player roles.
 6. The method of claim 1, wherein the request comprises substituting a player that was in a play of a historical game with a target player that was not in the play.
 7. A non-transitory computer readable medium comprising one or more sequences of instructions, which, when executed by a processor, causes a computing system to perform operations comprising: retrieving historical event data for a plurality of sporting matches, wherein retrieving the historical event data comprises: receiving broadcast video data corresponding to the plurality of sporting matches; deriving player tracking data from the broadcast video data; augmenting the player tracking data with the historical event data; generating one or more attributes of a team and of at least one player on the team based on the historical event data; receiving a request to simulate and view an alternative historical game outcome; simulating the alternative historical game outcome based on the one or more attributes generated by the computing system; and outputting a graphical representation of the simulation.
 8. The non-transitory computer readable medium of claim 7, the operations further comprising: generating body pose data for the at least one player based on the historical event data.
 9. The non-transitory computer readable medium of claim 8, wherein the simulating the alternative historical game outcome is based on the body pose data.
 10. The non-transitory computer readable medium of claim 7, wherein the graphical representation includes at least one skeletal representation of the at least one player, and wherein the at least one skeletal representation is based on the one or more attributes.
 11. The non-transitory computer readable medium of claim 7, wherein the one or more attributes include one or more player trajectories, one or more team styles, and one or more player roles.
 12. The non-transitory computer readable medium of claim 7, wherein the request comprises substituting a player that was in a play of a historical game with a target player that was not in the play.
 13. A system comprising: a processor; and a memory having programming instructions stored thereon, which, when executed by the processor, causes the system to perform operations comprising: retrieving, by a computing system, historical event data for a plurality of sporting matches, wherein retrieving the historical event data comprises: receiving, by the computing system, broadcast video data corresponding to the plurality of sporting matches; deriving, by the computing system, player tracking data from the broadcast video data; augmenting, by the computing system, the player tracking data with the historical event data generating, by the computing system, one or more attributes of a team and of at least one player on the team based on the historical event data; receiving, by the computing system, a request to simulate and view an alternative historical game outcome; simulating, by the computing system, the alternative historical game outcome based on the one or more attributes generated by the computing system; and outputting, by the computing system, a graphical representation of the simulation.
 14. The system of claim 13, the operations further comprising: generating, by the computing system, body pose data for the at least one player based on the historical event data.
 15. The system of claim 14, wherein the simulating the alternative historical game outcome is based on the body pose data.
 16. The system of claim 13, wherein the graphical representation includes at least one skeletal representation of the at least one player, and wherein the at least one skeletal representation is based on the one or more attributes.
 17. The system of claim 13, wherein the one or more attributes include one or more player trajectories, one or more team styles, and one or more player roles.
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